

# Chuqin (Allen) Geng

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## Research interests

Mechanistic interpretability of large language models (LLMs); interpretable and reliable learning across graph, vision, and language; neuro-symbolic AI; neural network robustness and formal verification; AI for science.

## Education

**University of Toronto** Toronto, ON, CA  
Visiting scholar Jan 2023 – Present  
Advisor: Prof. Xujie Si

**McGill University, Mila - Quebec AI Institute** Montreal, QC, CA  
PhD in Computer Science Sep 2021 – Present  
Advisor: Prof. Xujie Si, *GPA: 4.0/4.0*

**Georgia Institute of Technology** Atlanta, GA, USA  
MSc in Computer Science, *GPA: 4.0/4.0* Sep 2019 – Apr 2021

**University of Reading** Reading, Berkshire, UK  
MSc in Finance, *GPA: 4.0/4.0* Sep 2016 – Dec 2017

**University of Toronto** Toronto, ON, CA  
Honours BSc in Math and Statistics Sep 2011 – Aug 2015

## Selected Papers

**Are We Recovering Mechanisms? Objective-Level Recovery Gaps in Mechanistic Interpretability**

**Chuqin Geng**, Li Zhang, Haolin Ye, Mark Zhang, Luke Zhang, Xujie Si  
*In submission*

**Rethinking Circuit Evaluation: Do Circuits Explain Model Errors?**

Li Zhang\*, **Chuqin Geng\***, Mark Zhang, Chen Yang, Luke Zhang, Haolin Ye, Xujie Si  
*In submission*

**VisionLogic: Discovering and Grounding Decision-Relevant Visual Concepts**

**Chuqin Geng**, Yuhe Jiang, Li Zhang, Zhaoyue Wang, Haolin Ye, Mark Zhang, Jingkai Xu, Xujie Si  
*In submission*

**Beyond Message Passing: A Symbolic Alternative for Expressive and Interpretable Graph Learning**

**Chuqin Geng**, Li Zhang, Haolin Ye, Ziyu Zhao, Yuhe Jiang, Tara Saba, Xinyu Wang, Xujie Si

*In submission*

**CROSSREF-VQA: Benchmarking Cross-modal Reference Resolution in Structured Graphics**

Yuhe Jiang, Youyuan Zhang, Hangrui Bi, **Chuqin Geng**, Xujie Si

*In submission*

**Neural Proposals, Symbolic Guarantees: Neuro-Symbolic Graph Generative Modeling**

**Chuqin Geng**, Li Zhang, Mark Zhang, Zhaoyue Wang, Haolin Ye, Xujie Si

*NeurIPS 2026*

**DreamProver: Evolving Transferable Lemma Libraries via a Wake-Sleep Theorem-Proving Agent**

Youyuan Zhang\*, Jialiang Sun\*, Hangrui Bi, **Chuqin Geng**, Wenjie Ma, Zhaoyu Li, Xujie Si

*COLM 2026*

**LogicXGNN: Grounded Logical Rules for Explaining Graph Neural Networks**

**Chuqin Geng**, Ziyu Zhao, Zhaoyue Wang, Haolin Ye, Yuhe Jiang, Xujie Si

*ICLR 2026 (Top ~2%)*

**NEUROLOGIC: From Neural Representations to Interpretable Logic Rules**

**Chuqin Geng**, Anqi Xing, Li Zhang, Ziyu Zhao, Yuhe Jiang, Xujie Si

*UCRL @ ICLR 2026*

**Learning Minimal Neural Specifications**

**Chuqin Geng**, Zhaoyue Wang, Haolin Ye, Xujie Si

*NeuS 2025 (Oral)*

**Towards Robust Saliency Maps**

Van Nham Le\*, **Chuqin Geng**\*, Xujie Si, Arie Gurfinkel

*ACML 2024*

**Towards Reliable Neural Specifications**

**Chuqin Geng**\*, Van Nham Le\*, Xiaojie Xu, Zhaoyue Wang, Arie Gurfinkel, Xujie Si

*ICML 2023 (Oral)*

**Identifying Different Student Clusters in Functional Programming Assignments: From Quick Learners to Struggling Students**

**Chuqin Geng**, Wenwen Xu, Yingjie Xu, Brigitte Pientka, Xujie Si

*SIGCSE 2023 TS*

**TorchProbe: Fuzzing Dynamic Deep Learning Compilers**

Qidong Su, **Chuqin Geng**, Gennady Pekhimenko, Xujie Si

*APLAS 2023*

**Scalar Invariant Networks with Zero Bias**

**Chuqin Geng**, Xiaojie Xu, Haolin Ye, Xujie Si

*NeuReps @ NeurIPS 2023*

**Novice Type Error Diagnosis with Natural Language Models**

**Chuqin Geng**, Haolin Ye, Yixuan Li, Tianyu Han, Brigitte Pientka, Xujie Si

*APLAS 2022*

**SAT-DIFF: A Tree Diffing Framework Using SAT Solver**

**Chuqin Geng**, Haolin Ye, Yihan Zhang, Brigitte Pientka, Xujie Si

*preprint, [arxiv.org/abs/2404.04731](https://arxiv.org/abs/2404.04731)*

**Can ChatGPT Pass An Introductory Level Functional Language Programming Course?**

**Chuqin Geng**, Yihan Zhang, Brigitte Pientka, Xujie Si

*preprint, [arxiv.org/abs/2305.02230](https://arxiv.org/abs/2305.02230)*

**Selected awards**

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| SOCS Grad Stimulus Initiative (McGill University)                 | 2025      |
| Mila Research Scholarship (Mila - Quebec AI Institute)            | 2023–2025 |
| Graduate Excellence Award (McGill University)                     | 2021–2025 |
| Faculty of Science Grad Supplement (McGill University)            | 2023      |
| Apple Scholars nomination (McGill University)                     | 2023      |
| Computer Science Top-up Award (McGill University)                 | 2022      |
| Academic Distinction Award (University of Reading)                | 2017      |
| Second prize in mathematics competition (Shaanxi province, China) | 2009      |

**Research  
Experience**

**Explainable AI, Mechanistic Interpretability, and AI for Science**

Advisor: Prof. Xujie Si (University of Toronto) Jan 2025 – Present

Developed grounded logical explanations for graph neural networks, revealing molecular substructures associated with their predictions. Combined neural graph generation with symbolic solving to generate molecules satisfying hard constraints, enabling controllable design for drug and materials discovery. Identified limitations in current mechanistic interpretability methods for large language models.

### **Neural Network Verification, Robustness, and AI Safety**

Advisor: Prof. Xujie Si (University of Toronto) *Sep 2022 – May 2025*

Proposed neural activation patterns as verification specifications, extending certifiable input regions beyond hand-chosen local neighborhoods. Developed minimal neural specifications for concise, interpretable verification. Explored methods to improve model robustness, fairness, and safety.

### **Computer science education research**

*Sep 2021 – Dec 2022*

Advisors: Profs. Xujie Si and Brigitte Pientka (McGill University)

Studied transformer-based language models for diagnosing type errors made by novice programmers and conducted education research using statistical analysis of student activity logs, with the goal of auditing and improving the student learning experience.

### **Multimodal clinical prediction and reinforcement learning**

Graduate research (Georgia Institute of Technology) *Sep 2019 – Apr 2021*

Conducted research projects on multimodal mortality prediction using MIMIC clinical data and reinforcement learning for robot control.

### **Disproof of a conjecture in biometric security optimization**

Mentors: Prof. Steven Rayan (University of Toronto) *Jan 2015 – Aug 2015*

Disproved a conjecture regarding optimal solutions for biometric privacy-security trade-offs, providing new bounds for secure system optimization.

## **Teaching experience**

### **Head teaching assistant, McGill University**

*Winter 2022*

COMP 302: Programming Languages and Paradigms

Conducted weekly office hours and tutorials, designed and graded exams, developed auto-graders for assignments, implemented mutation testing, and utilized Moss for plagiarism detection.

## **Industry experience**

### **Cirqvia (startup project)**

Founder

*May 2026 – Present*

Applying interpretability research to practical AI alignment, safety, and reliability needs; received early-stage funding.

### **FITFI Inc.**

Toronto, CA

Senior Data Scientist

*Jan 2018 – Sep 2019*

Developed a system for automatically detecting and monitoring exercise-equipment use. Led the algorithm team, secured a demo opportunity at the 2019 Collision Conference, and helped raise over 2 million CAD in funding.

### **SHAREWIN SOFTWARE**

Beijing, China

Algorithm Engineer

*Sep 2015 – Sep 2016*

Designed fault extraction algorithm with preprocessing, filtering, and ant tracking for accurate fault surface detection in 3D-seismic volumes.

## Talks

**Learning Minimal Neural Specifications** *May 2025*  
NeuS 2025

**Towards Reliable Neural Specifications** *Aug 2023*  
ICML 2023

**A study on student performance clusters** *Mar 2023*  
SIGCSE 2023, SPLICE Workshop

**Novice Type Error Diagnosis with Natural Language Models** *Dec 2022*  
APLAS 2022

## Service

I contribute to academic service through conference reviewing and student selection:

- **Machine Learning:** ICML (2023–2026), ICLR (2024–2027), NeurIPS (2023–2026), AAAI (2024–2027)
- **Computer Vision:** CVPR (2024, 2025), ECCV 2026
- **Formal Methods:** TACAS 2025
- **HCI & Education:** SIGCSE 2023, CHI 2023
- **Supervision Evaluation Committee Member:** Mila Supervision Request Process 2027

## Mentoring

I have mentored students from the following institutions:

- **University of Toronto:** Li Zhang, Mark Zhang
- **McGill University:** Ziyu Zhao, Haolin Ye, Yihan Zhang, Zhaoyue Wang, Xiaojie Xu
- **McMaster University:** Jingkai Xu
- **Tsinghua University:** Chen Yang